

PROFESSIONAL SUMMARY

Industrial Engineering Ph.D. candidate with 4+ years of experience building and validating **discrete-event simulation, digital-twin, and optimization models** for automated production systems. Model end-to-end material flow across equipment, conveyors, queues, and WIP to support **capacity planning, bottleneck analysis, line balancing, and operating-policy decisions**. Hands-on experience connecting models to physical equipment and imperfect operating data using Python, PLCnext, OPC UA.

EDUCATION

The Pennsylvania State University *University Park, PA*

Ph.D. in Industrial Engineering

Expected: Dec 2026

President, Society of Manufacturing Engineers (SME) Penn State Chapter

Rising Star, ASME Dynamic Systems and Control Division (DSCD), 2025

Tianjin University *Tianjin, China*

B.S. & M.S. in Civil Engineering

May 2017 & Jan 2020

Exchange: Nanyang Technological University, Singapore (2019–2020)

RESEARCH & PROFESSIONAL EXPERIENCE

Manufacturing Systems Engineering Intern *New Kensington, PA*

Digital Foundry at New Kensington

Jan 2026 – Present

- Develop the **control architecture and system design** for Iron MaBLLe, a reconfigurable automated bottling line with cap-fill, inspection, pasteurization, and labeling workstations connected by a conveyor.
- Map multi-stage material flow, buffers, equipment interactions, and cycle-time constraints to identify bottlenecks and evaluate line configurations, control policies, and production performance.
- Conduct stepped-frequency conveyor tests, pasteurizer heating/cooling experiments, and factorial throughput studies; collect **5-Hz equipment data** through PLCnext, OPC UA, and Ignition to characterize inputs and validate system models.

Discrete-Event Simulation & Digital-Twin Decision Support

Penn State — Advisors: Prof. Kovalenko & Prof. Pangborn

Aug 2022 – Present

- Built **Python/MATLAB event-driven simulation and digital-twin models** of end-to-end manufacturing flow, representing machines, buffers, WIP, inventories, transportation links, and discrete-batch operations [[Demo](#)].
- Conducted **what-if studies** of routing, staging, buffer levels, bottleneck conditions, operating policies, production timing, and resource utilization to support capacity and system-design decisions.
- Calibrated process rates and machine dynamics using physical experiments and time-stamped operating data; validated model predictions and control recommendations on the Iron MaBLLe testbed through PLC-based execution.

Production Systems Optimization & Control

Penn State — Advisors: Prof. Kovalenko & Prof. Pangborn

Aug 2022 – Present

- Formulated **mixed-integer optimization** and model predictive control algorithms for multi-stage production scheduling, inventory management, and throughput coordination under capacity, timing, and shared-resource constraints.
- Quantified production, WIP/inventory, utilization, timing, and energy-cost tradeoffs under stochastic demand and utility-price scenarios, translating model results into implementable operating schedules.
- Reduced energy cost by **28.2%** through optimized production schedules validated on a physical bottling-line testbed using real utility pricing data.

Research Associate, Forecasting & Operations Analytics

Surbana Jurong–NTU Corporate Lab, Singapore

Jan 2020 – Aug 2021

- Developed Bayesian time-series and data-driven forecasting models from noisy, high-dimensional operational data to characterize uncertainty and support resource-planning decisions.
- Partnered with engineering, operations, and finance stakeholders to communicate assumptions, uncertainty, and decision-ready recommendations for system-performance improvement.

SELECTED PUBLICATIONS

Li, H., Pangborn, H. C., & Kovalenko, I. (2025). Bi-level Model Predictive Control for Energy-aware Integrated Product Pricing and Production Scheduling. *ASME Modeling, Estimation and Control Conference (MECC)*. *Best Student Paper Finalist*.

Li, H., Pangborn, H. C., & Kovalenko, I. (2025). Hierarchical Model Predictive Control for Energy-aware Scheduling of Digital Twin-based Batch Manufacturing Systems. *IEEE Transactions on Automation Science and Engineering*.

TECHNICAL SKILLS

Simulation & Optimization: Discrete-Event Simulation, Digital Twins, State-Space Modeling, Mixed-Integer Programming, Stochastic Programming, Model Predictive Control, FlexSim, SimEvent, SimPy

Programming & Data: Python, SQL, MATLAB, NumPy, Pandas, PyTorch, Excel, Gurobi, YALMIP, Git, Linux

Automation & Design: PLCnext, OPC UA, Ignition, IEC 61131-3, AutoCAD, SolidWorks